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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR /IMPROVEMENT/ REAPPEARANCE EXAMINATION NOVEMBER 2025

FIRST SEMESTER

Discipline Specific Core Course (DSC) - MG1DSCFIM101 - GENERAL CONDITIONING AND RECREATION

(2024 ADMISSION ONWARDS)

Answer key

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer **All** Questions

Each Question carries **1** mark

1. Monitoring target heart rate during exercise helps to: [U] / [CO2]

   a). Prevent overtraining

2. In calisthenics, a “burpee” combines which two movements? [K] / [CO1]

   c). Squat and jump

3. Pull-ups primarily develop which of the following muscle groups? [K] / [CO3]

   b). Latissimus dorsi and biceps

4. In a hang power snatch, the bar is lifted from [K] / [CO4]

b). Mid-thigh or knee height to overhead

5. Which of the following is an example of HIIT? [U] / [CO2]

b). 30 seconds sprint + 90 seconds walk, repeated

6. Which of these benefits is most associated with regular slow running? [K] / [CO1]

b). Improved cardiovascular health and endurance

7. What is one of the primary benefits of swimming? [U] / [CO2]

b). It improves cardiovascular fitness and overall muscle tone

8. General conditioning primarily focuses on: [K] / [CO2]

b). Overall development of physical fitness

9. What type of rope is typically used in skipping workouts? [U] / [CO2]

b). Light and adjustable jump rope

10. The Iron Man exercise is also known as: [U] / [CO3]

c). Prone back extension hold

[10x1 = 10]

Part B

Short Answer Type Questions

Answer any **10** Questions

Each Question carries **2** marks

11. What is the primary goal of repetitive training? [U] / [CO1]

To improve skill efficiency, muscle memory, and performance consistency by repeatedly practicing a specific movement or activity.

12. Differentiate between agility and speed with examples [A] / [CO1]

Speed: Ability to move quickly in one direction. Example: 100-meter sprint.

Agility: Ability to change direction quickly while maintaining control. Example: Dodging opponents in football.

13. Why are rope workouts effective for metabolic conditioning? [U] / [CO1, CO2]

Because they involve continuous high-intensity movements, increasing heart rate, enhancing calorie burn, and improving both anaerobic and aerobic capacity.

14. Which muscles are primarily involved in the sit-up portion? [U] / [CO4]

Rectus abdominis

Hip flexors (iliopsoas)

Obliques (as stabilizers)

15. Differentiate between static and dynamic stretching [K] / [CO4]

Static stretching: Holding a stretch for 15–30 sec without movement. Example: Hamstring hold.

Dynamic stretching: Stretching while moving through range of motion. Example: Leg swings.

16. What is cycling? [K] / [CO1]

A cardiovascular exercise performed on a bicycle or stationary cycle to improve endurance, leg strength, and overall fitness.

17. Explain the importance of warm-up before lifting [U] / [CO2]

Warm-up:

Increases blood flow to muscles

Enhances joint mobility

Reduces risk of injury

Improves performance by activating the nervous system

18. What is aerobics? [K] / [CO1]

Aerobics is rhythmic, continuous exercise that uses large muscle groups to improve cardiovascular endurance, such as jogging, dancing, or step aerobics.

19. Mention two core muscles that play a major role in agility training [K] / [CO1]

1.Rectus abdominis

2.Obliques (Internal or External)

20. Name two factors affecting heart rate response during exercise. [U] / [CO1]

1.Intensity of exercise

2.fitness level

21. What type of fitness component does the Bird Dog mainly improve? [U] / [CO4]

Core stability and balance

Also improves spinal alignment and posture

22. How is an anaerobic and aerobic workout combined with Fartlek training? [U] / [CO1]

Fartlek alternates fast (anaerobic) bursts with slow/moderate (aerobic) running, creating a mixed- intensity endurance workout.

[10x2 = 20]

Part C

Short Essay Type Questions

Answer any **4** Questions

Each Question carries **5** marks

23. how the FITT formula can be individualized based on age, fitness level, and training goals? [K] / [CO2]

FITT = Frequency, Intensity, Time, Type

Age: Older individuals require lower intensity and longer recovery.

Fitness level: Beginners start with lower frequency and time; advanced athletes train longer and harder.

Training goals:

Weight loss → higher frequency, moderate intensity

Strength → high intensity, lower reps

Endurance → longer duration, moderate intensity

FITT ensures a personalized, safe, and progressive training plan.

24. How can running long slow distances (LSD) help the development of aerobic capacity? [K] / [CO1]

LSD training:

Increases stroke volume of the heart
Enhances oxygen delivery to muscles
Improves mitochondrial density
Strengthens slow-twitch muscle fibers
Builds endurance with low injury risk
This results in greater aerobic capacity (VO_2 max) and stamina.

25. Analyze the biomechanical movements involved in performing a Burpee? [A] / [CO3]

A burpee includes several coordinated actions:

1. Squat: Hip and knee flexion; quads and glutes engaged.
2. Plank position: Shoulder stabilization; core tight.
3. Push-up: Elbow flexion/extension; chest, triceps, shoulders work.
4. Return to squat: Hip and knee extension.
5. Vertical jump: Explosive power using calves, quads, glutes.

It is a full-body movement combining strength, power, coordination, and cardiovascular endurance

Here are the answers to the questions shown in your image:

26. Explain how regular flexibility and balance training enhance efficiency of movement and prevent musculoskeletal injuries?

Regular flexibility and balance training improves the functional capacity of the musculoskeletal and neuromuscular systems.

How Flexibility Helps

1. Increases Range of Motion (ROM): Flexible muscles and joints move more freely, allowing smoother and more efficient movement.

2. Reduces Muscle Stiffness: Decreases tension in muscles, lowering the risk of strains and sprains.

3. Enhances Posture and Alignment: Proper joint alignment reduces unnecessary stress on ligaments and tendons.

4. Improves Functional Movement: Better ROM allows athletes to execute skills more efficiently and with less energy expenditure.

How Balance Training Helps

1. Strengthens Stabilizing Muscles: Activates core and small stabilizers that support joints.

2. Improves Proprioception: Enhances awareness of body position, reducing chances of falls or missteps.

3. Prevents Overuse Injuries: Balanced muscle activation prevents excessive load on specific muscles or joints.

4. Enhances Coordination: Better neuromuscular control supports smooth and safe movement patterns.

Overall Effect

Together, flexibility and balance training increase movement efficiency, reduce joint stress, and significantly lower the likelihood of musculoskeletal injuries

27. Discuss how percentage of HRmax can be used to plan training for endurance athletes?

Heart Rate Maximum (HRmax) is a key tool to monitor exercise intensity and structure endurance training zones.

Steps to Use HRmax

1. Estimate HRmax:

Common formula: HRmax = 220 – age (or more accurate lab tests).

2. Determine Training Zones (% of HRmax):

Zone 1 (50–60% HRmax): Light intensity—warm-up, cool-down, recovery.

Zone 2 (60–70% HRmax): Aerobic base training—improves fat metabolism.

Zone 3 (70–80% HRmax): Moderate intensity—enhances aerobic endurance.

Zone 4 (80–90% HRmax): High intensity—improves VO_2max and lactate threshold.

Zone 5 (90–100% HRmax): Maximum effort—short intervals for speed and power.

3. Match Training Goals to Zones:

Long-distance endurance athletes spend more time in Zone 2 and 3.

Interval training uses Zones 4 and 5 to increase speed and oxygen capacity.

Recovery sessions use Zone 1 to promote healing.

Benefits

Ensures safe and precise intensity control.

Prevents overtraining or undertraining.

Helps monitor progress and optimize performance

28. What role does Zumba play in enhancing cardiovascular endurance, and how does it compare to traditional aerobic exercises?

Zumba is a dance-based aerobic workout that uses rhythmic full-body movements, helping improve cardiovascular health.

Role of Zumba in Cardiovascular Endurance

1. Sustained Aerobic Activity: Continuous movement keeps heart rate in moderate to high zones.

2. High Energy Expenditure: Combines fast-paced music with dynamic motions, improving stamina.

3. Engages Multiple Muscle Groups: Full-body movement boosts overall cardiovascular efficiency.

4. Improves Adherence: Fun, social nature increases consistency, which is key for endurance improvement.

Comparison with Traditional Aerobics

Aspect Zumba Traditional Aerobics

Intensity Moderate to high, varies with choreography More controlled and predictable

Engagement Fun, dance-based, motivating structured exercises, can feel repetitive

Calorie Burn High due to continuous movement Moderate to high depending on routine

Skill Requirement Simple dance steps; enjoyable basic aerobic steps

Effectiveness Equally effective for cardiovascular endurance Conclusion Proven to improve aerobic capacity